Inferential Feature Analysis: Latino Trump Support (CMPS 2016)

Analysis Pipeline — Random Forest + SHAP + Bootstrap

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Table 1: Analysis Configuration.

Parameter	Value
Bootstrap iterations (B)	500
Trees per forest	500
Test proportion	0.25
Rank stability threshold (k)	3
Development mode	FALSE
Raw data file	38040-0001-Data.rda

Table 2: Latino Subsample Summary.

Item	Value
Latino N	3002
Trump voters	594
Trump %	19.8%
Weight range	[0.06, 9.73]

Table 3: Train/Test Split Summary.

Set	N	Trump %	Features
Training	2252	19.9	862
Test	750	19.3	862

Table 4: Progressive Exclusion Tiers.

Tier	Train N	Test N	Features	Excluded
1: Full	2252	750	862	0
2: Non-Tautological	2252	750	849	13
3: Non-Partisan (Ring 1)	2252	750	831	31

Table 5: Tier 3 AUC Across Learners.

Learner	Test AUC
Random Forest	0.861
Elastic Net Logistic	0.897
XGBoost	0.920

1 Analysis Parameters

2 Data and Sample Construction

3 Preprocessing

4 Train/Test Split and Imputation

5 Tier Construction and Partisan Leakage Audit

6 Model Fitting

7 Benchmark Models

Elastic net logistic regression and XGBoost are fitted on the same Tier 3 train/test split to confirm that findings are a property of the data rather than an artifact of Random Forest.

8 SHAP Computation

9 Feature Correlation Structure

When features are correlated, SHAP distributes predictive credit across correlated items in ways that are mathematically principled but may not map onto standalone constructs. The correlation matrix below documents the pairwise relationships among the top features.

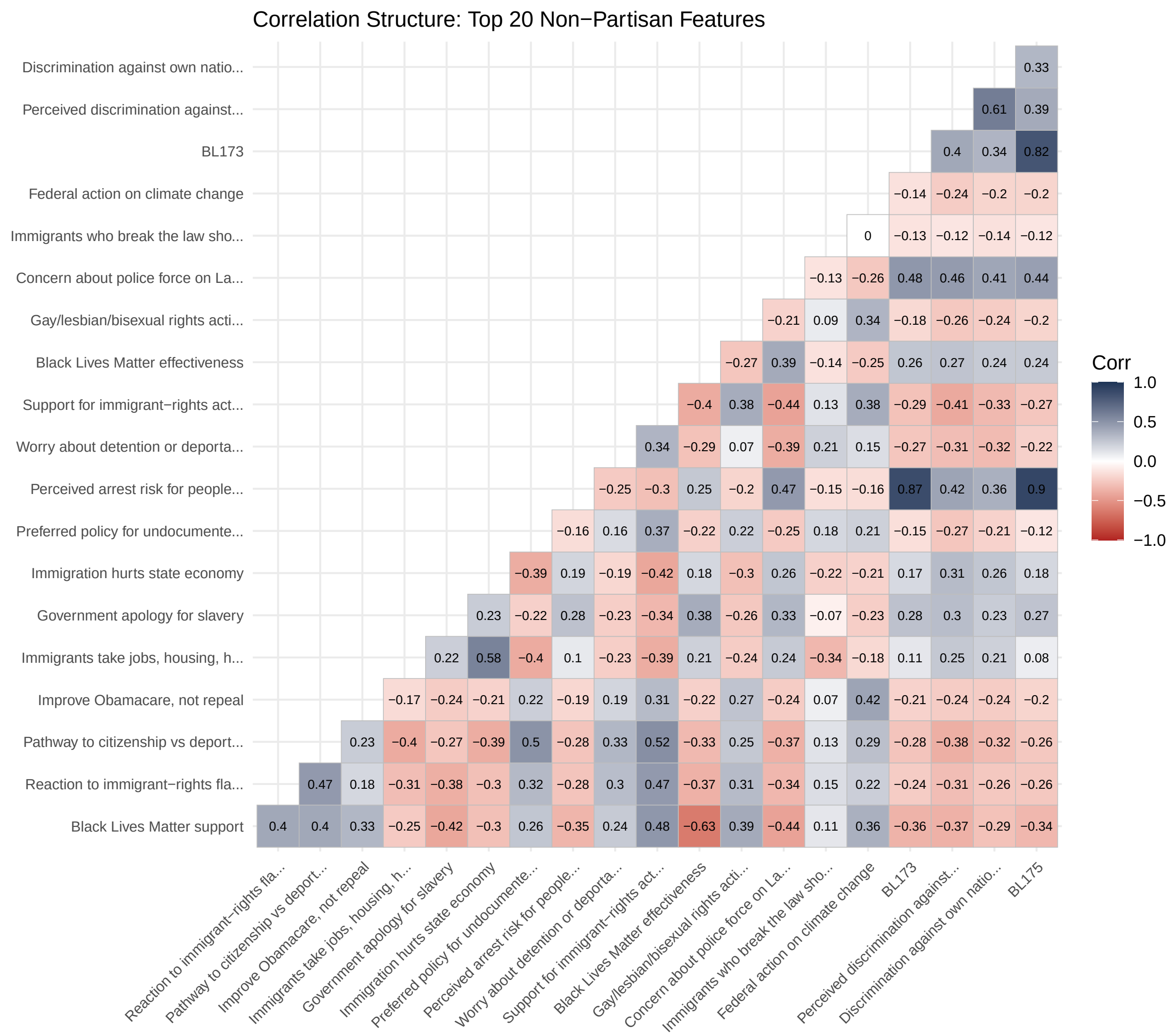

Figure 1: Pairwise Pearson correlations among top 20 Tier 3 features (test set).

10 Bootstrap Confidence Intervals

The bootstrap resamples the training set 500 times, refitting a Random Forest and computing TreeSHAP on the fixed test set each time. This isolates model instability from evaluation-set composition variance. Each feature's 95% CI is the 2.5th–97.5th percentile of the bootstrap distribution of $E_w[|\phi_j|]$.

Table 6: Threshold Sensitivity: Stable Features at k = 2.0, 3.0, and 4.0.

Rank	Feature	sigma_j	k=2.0	k=3.0	k=4.0
1	Black Lives Matter support	1.48			
2	Reaction to immigrant-rights flag displays	2.15			
3	Pathway to citizenship vs deportation	2.19			
4	Improve Obamacare, not repeal	1.93			
5	Immigrants take jobs, housing, health care	2.75			
6	Government apology for slavery	2.75			
7	Immigration hurts state economy	4.29			
8	Preferred policy for undocumented immigrants	4.90			
9	Perceived arrest risk for people like me	6.45			
10	Worry about detention or deportation	8.19			
11	Support for immigrant-rights activism	6.08			
12	Black Lives Matter effectiveness	5.94			
13	Gay/lesbian/bisexual rights activism	8.96			
14	Concern about police force on Latinos	7.74			
15	Immigrants who break the law should leave	10.84			
16	Federal action on climate change	8.26			
17	BL173	8.10			
18	Perceived discrimination against Latinos	6.93			
19	Discrimination against own national-origin subgroup	9.40			
20	BL175	9.79			

Table 7: Spearman Rank Correlations Across 5 Independent Splits (min = 0.857, mean = 0.857).

Split	Split_1	Split_2
Split_1	1.000	0.857
Split_2	0.857	1.000

11 Results Assembly

12 Threshold Sensitivity

The stability threshold k is a pre-specified reporting rule, not a universal standard. The table below shows which features qualify as stable findings at $k = 2.0$, $k = 3.0$, and $k = 4.0$, holding the magnitude criterion (ci_lo > 0) constant throughout.

13 Repeated-Resampling Robustness

The bootstrap assesses training-sample sensitivity conditional on one fixed test partition. To confirm the feature architecture is not an artifact of the particular split, five independent 75/25 partitions are drawn and the full Tier 3 RF + SHAP pipeline is run on each. Spearman rank correlations of feature importance across all pairs test whether the same features surface regardless of the split.

Table 8: AUC Across the Progressive Exclusion Framework.

Model	Test AUC	Drop from T1	Boot Mean AUC	95% CI
Tier 1 (Full)	0.949	NA	NA	—
Tier 2 (Non-Tautological)	0.913	0.035	NA	—
Tier 3 (Non-Partisan)	0.861	0.088	0.847	[0.832, 0.862]

Table 9: Stable Ring 1 Non-Partisan Features (ci_lo > 0, sigma_j <= 3).

Rank	Feature	Code	E_w[|phi|]	CI lo	CI hi	sigma_j
1	Black Lives Matter support	C228	0.01230	0.00753	0.01456	1.48
2	Reaction to immigrant-rights flag displays	C142	0.01177	0.00582	0.01373	2.15
3	Pathway to citizenship vs deportation	C141	0.01143	0.00590	0.01432	2.19
4	Improve Obamacare, not repeal	C45	0.01079	0.00619	0.01304	1.93
5	Immigrants take jobs, housing, health care	BLA207	0.00872	0.00472	0.01215	2.75
6	Government apology for slavery	C158	0.00723	0.00464	0.01201	2.75

14 Main Results

14.1 Model Performance

14.2 Stable Non-Partisan Features

Features are reported as stable findings under a conjunctive decision rule pre-specified before examining results: the bootstrap 95% CI lower bound for $E_w[|\phi_j|]$ must exceed zero (magnitude criterion) AND $\sigma_j \leq 3$ (rank stability criterion). The first condition ensures the feature's importance is bounded away from nothing across 500 training resamples. The second ensures its ordinal position in the importance ranking is stable enough to support comparative claims.

14.3 Figure 1. Feature Importance (Beeswarm)

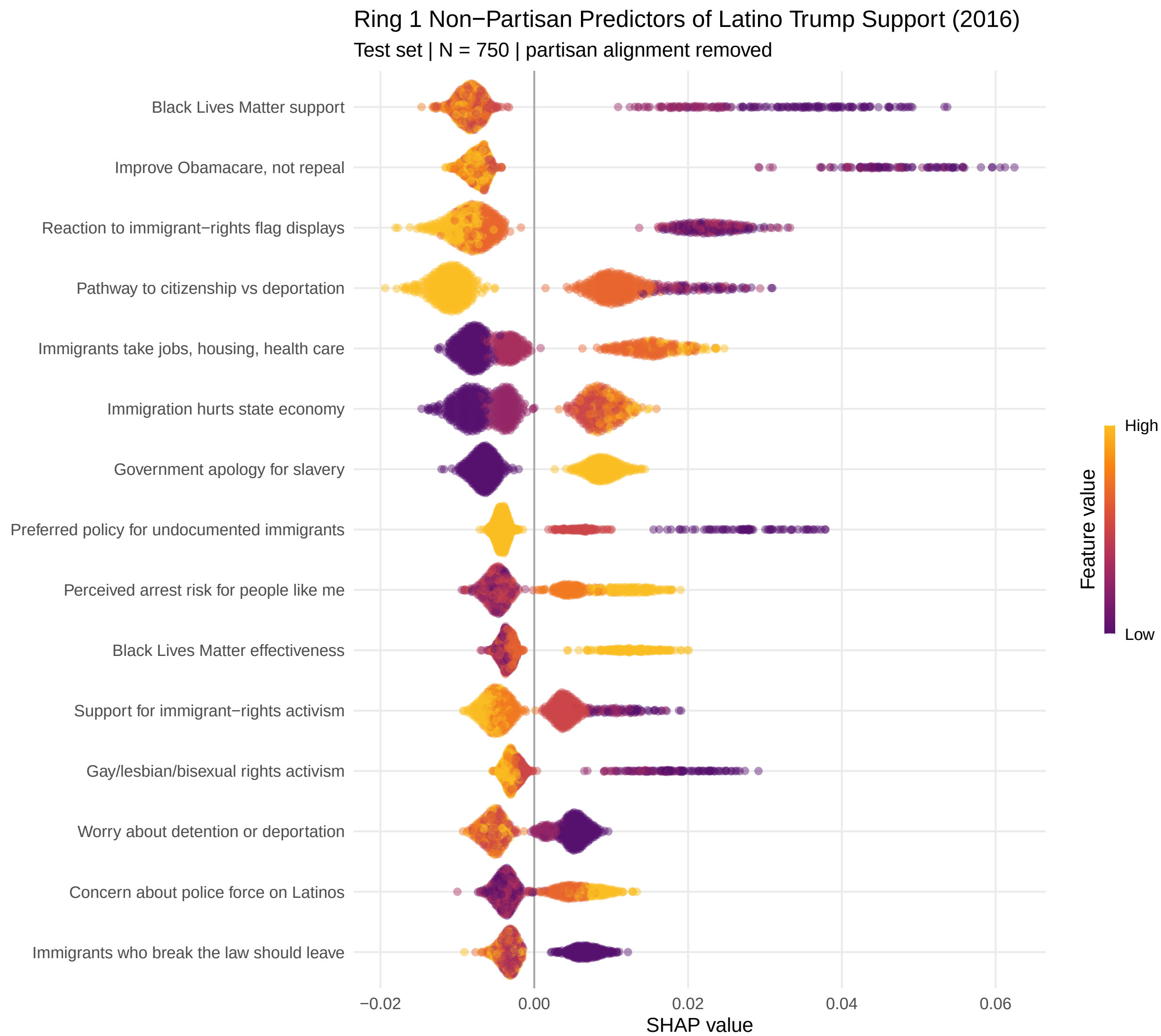

Figure 2

14.4 Figure 2. Bootstrap Confidence Intervals

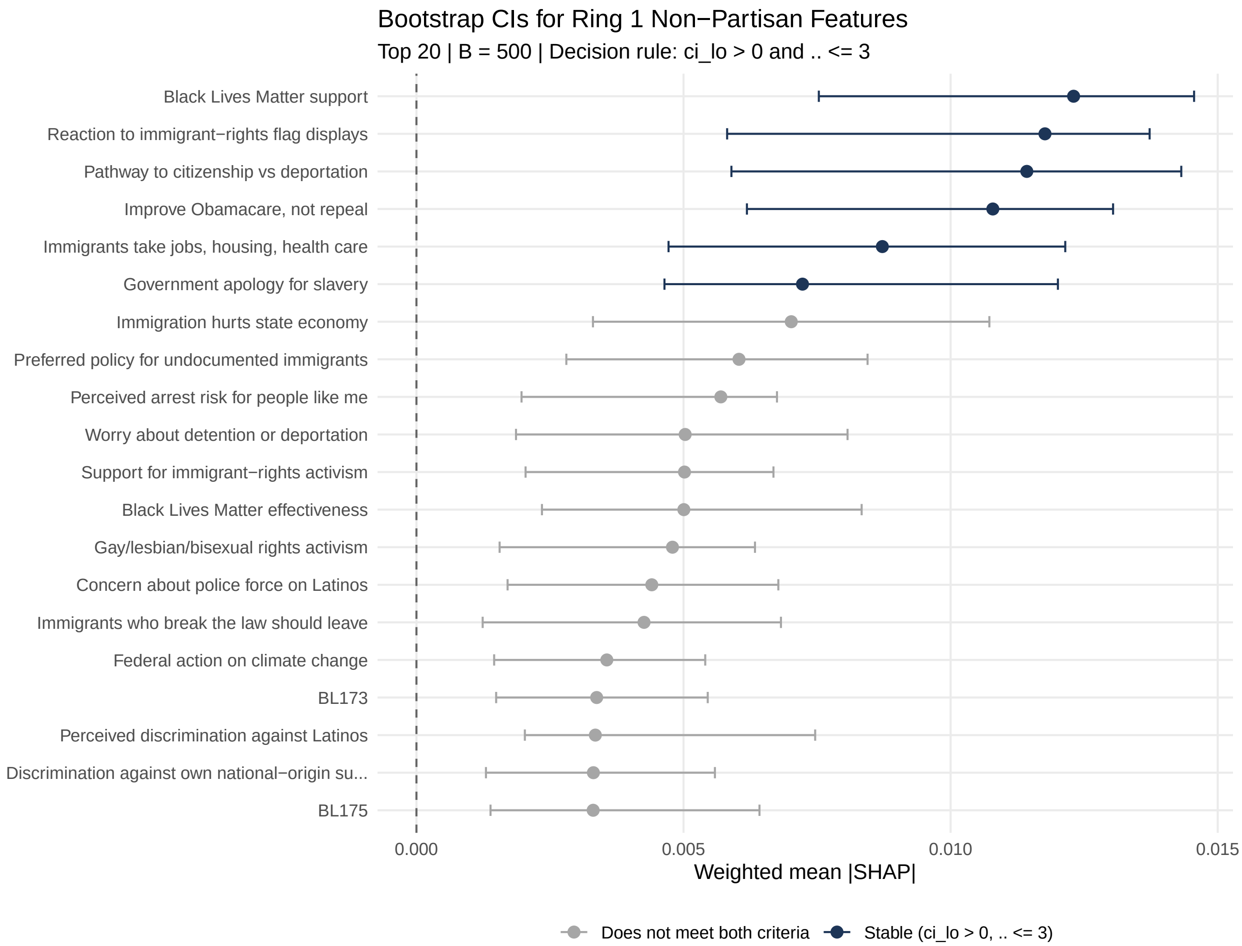

Figure 3

14.5 Figure 3. SHAP Dependence (Top Feature)

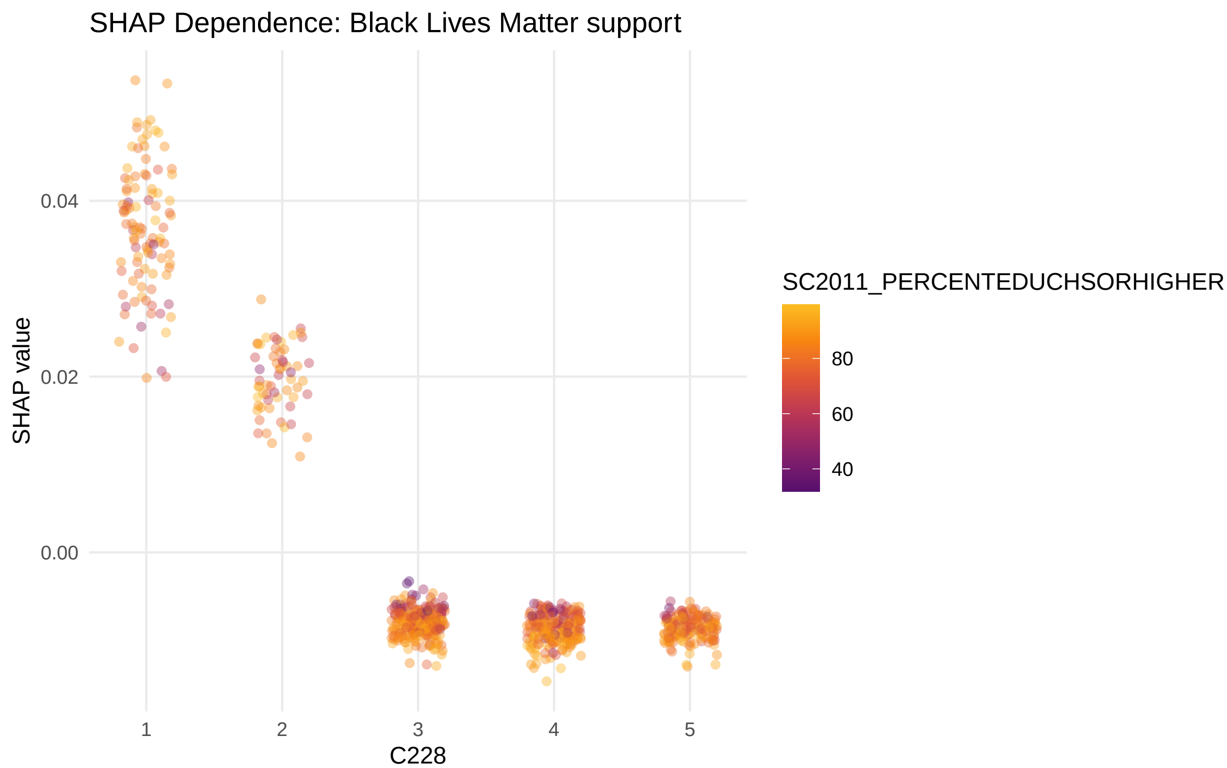

Figure 4

15 Save Results Bundle

All model objects, SHAP matrices, bootstrap results, tier metadata, and summary tables are saved to a single `.rds` file. The reporting QMD loads this file to generate all manuscript figures and tables without re-running the model.

The results bundle has been saved to `ifa_results.rds` (32.1 MB). Load this file in the reporting QMD with `readRDS("ifa_results.rds")`.

16 Session Information

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